Week 3b - Quiz

07 May 2025 13:59

A connecting rod in a high-performance internal combustion engine is a critical component. If it fails then the engine fails. To minimise inertial forces and bearing loads it must weigh as little as possible. What are the best materials for the manufacture of con rods?

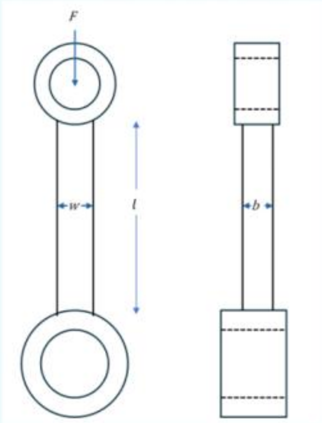
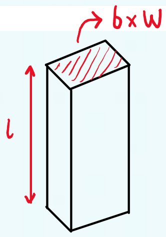


Fig 1. A connecting rod that must not buckle under elastic loading.

The conditions that are optimised require that the best material for a connecting rod would be the one with the minimum weight that will carry a peak force F without buckling. For simplicity, we will assume that the shaft is of length l and has a rectangular uniform cross-section of area A, which is equal to $b \times w$.

What is the volume of the conrod?



$\therefore V = l \times b \times w = lbw$

- a. bwl
- b. πAl
- c. $Abwl$
- d. Al/π

Ignoring the cylinders at the end what is the mass, m, of the central section of the conrod?

$m = \rho V = \rho lwb$

- a. $m = Al\rho/\pi$
- b. $m = \pi Al\rho$
- c. $m = Abwl\rho$
- d. $m = bwl\rho$

Assuming that the force, F, that can be applied to the conrod must be maintained at less than the Eulerian bucking load which is given as $F \leq \frac{\pi^2 EI}{l^2}$ and also knowing that for a rectangular cross-section that $I = \frac{bw^3}{12}$, calculate the maximum force prior to buckling in the conrod.

$$F \leq \frac{\pi^2 E\left(\frac{bw^3}{12}\right)}{l^2} = \frac{\pi^2 E b w^3}{12 l^2}$$

- a. $F \leq \frac{\pi^4 Ebw^2}{12l^4}$
- b. $F \leq \frac{\pi^2 Ebw^2}{12l^2}$
- c. $F \leq \frac{\pi^2 Ebw^3}{12l^2}$
- d. $F \leq \frac{\pi^3 Ebw^3}{12l^3}$

The buckling could take place in either the *b* or the *w* axis depending upon the specific end conditions. For simplicity, it is assumed that the frequency for each mode should be equal and so we have an additional constraint that the width, *w* is the same as the breadth *b*. So rearrange your equation from the previous question for the bucking load to make the area the subject, which of the following is the correct value for *A*?

$b = w$

$\therefore b^2 = w^2 = A$

- a. $A \geq \frac{l^2}{\pi}\frac{\sqrt{12F}}{E}$ ✗
- b. $A \geq \frac{l}{\pi}\frac{\sqrt{12F}}{E}$ ✗
- c. $A \geq \frac{l}{\pi}\frac{\sqrt{12F}}{\sqrt{E}}$ ✓
- d. $A \geq \frac{l^2}{\pi}\frac{\sqrt{12F}}{\sqrt{E}}$ ✗

$$F \leq \frac{\pi^2 E A^2}{12 l^2}$$

$$A \geq \sqrt{\frac{12 l^2 F}{\pi^2 E}} = \sqrt{\frac{12F}{E}}\,\frac{l}{\pi}$$

Substitute the equation for the area, *A*, back into the mass equation given in the above question to derive a new equation for the mass of the conrod.

- a. $m \geq \frac{\sqrt{12F}}{\pi} l^3 (\frac{\rho}{E})$ ✗
- b. $m \geq \frac{\sqrt{12F}}{\pi} l^2 (\frac{\rho}{\sqrt{E}})$ ✓
- c. $m \geq \frac{\sqrt{12F}}{\pi} l^2 (\frac{\rho}{E})$ ✗
- d. $m \geq \frac{\sqrt{12F}}{\pi} l^3 (\frac{\rho}{\sqrt{E}})$ ✗

$$m = \rho l w b = \rho l A = \frac{\rho l^2}{\pi}\sqrt{\frac{12F}{E}} = \frac{\sqrt{12F}}{\pi} l^2 \left(\frac{\rho}{\sqrt{E}}\right)$$

The materials index that generates the best materials for use in a conrod application would therefore be the one that **maximises** which of the following materials index values?

Minimise mass : $m \downarrow$

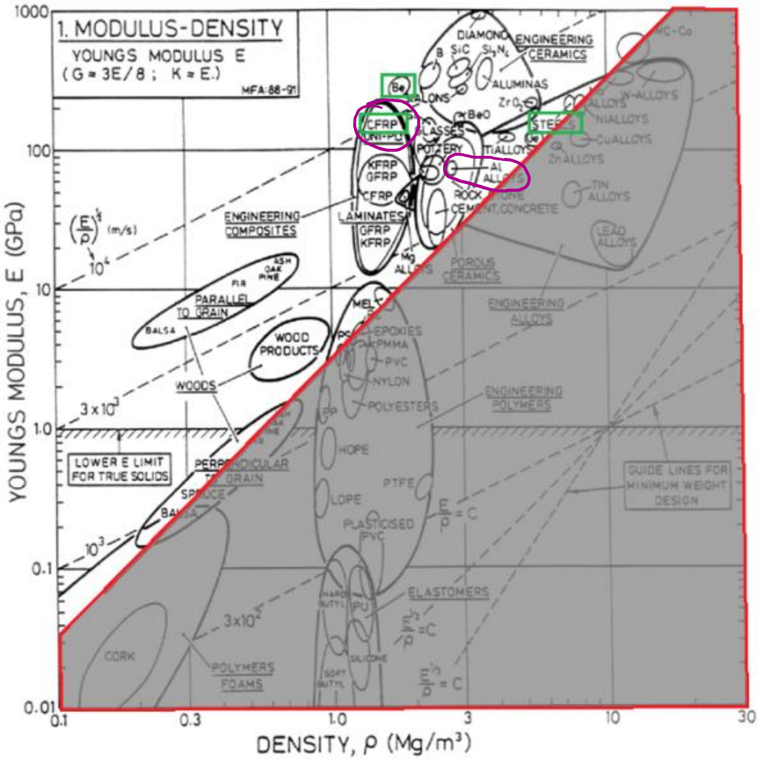
- a. $M = \frac{E^2}{\rho}$ ✗
- b. $M = \frac{\sqrt{E}}{\rho}$ ✓
- c. $M = \rho.\sqrt{E}$ ✗
- d. $M = \frac{\rho}{\sqrt{E}}$ ✗

$\therefore \left(\frac{\rho}{\sqrt{E}}\right)\downarrow$ minimised or $\left(\frac{\sqrt{E}}{\rho}\right)\uparrow$ maximised

Which of the following units is most appropriate for the materials index you have identified in the previous question?

- a. $GPa^{-0.5}\ Mg/m^3$ ✗
- b. $GPa^{0.5}\ Mg/m^3$ ✗
- c. $GPa^2\ m^3/Mg$ ✗
- d. $GPa^{0.5}\ m^3/Mg$ ✓

$$\frac{\sqrt{GPa}}{Mg/m^3} = GPa^{0.5}\ m^3/Mg$$



— For next question

Using the list of materials below identify the two most suitable materials that can be used to make a high-performance conrod.

- a. Tin alloys
- ✓ b. Carbon fibre composites
- c. Lead alloys
- ✓ d. Aluminium alloys
- e. Cork
- f. Silicone Rubber
- g. Nylon
- h. Zinc alloys
- i. Cast iron

Look at materials above the index line plotted.